

PHYS 357: Honours Quantum Physics 1

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Abstract

NOT FINISHED

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1 Introduction

Textbook used is JS Townsend A Modern Approach to Quantum Mechanics, 2nd ed., University Science Books 2012. Taught by Bill Coish. Mondays, Wednesdays and Fridays from 9:35 to 10:25 in Steward Biology Building N2/2. Course grading is divided as follows:

- 10% problem sets (must be handwritten scans).
- 40% quizzes (on wednesdays, best 8 of 12).
- 20% midterms (best of 2, october 2nd and november 6th).
- 30% final exam.

2 Stern-Gerlach Experiments

2.1 Spin

Definition 1 (Orbital Angular Momentum). *The orbital angular momentum $\vec{L}$ of a particle is defined as $\vec{L} = \vec{r} \times \vec{p}$, where $\vec{r}$ is the position vector and $\vec{p} = m \cdot \vec{v}$ is the linear momentum, both in the x - y plane.*

$L_z = rmv$ is the component of angular momentum along the z -axis (circular trajectory). For a solid object $L = I_0\omega$ where I_0 is the moment of inertia and ω is the angular velocity.

Periodic circular motion $I = \frac{q}{T}$, where T is the period.

A charged point particle of mass m and charge q moving in a circular path with period T will produce a circulating current $I = \frac{q}{T}$ and this current will produce a magnetic moment $\mu = \frac{IA}{c} = \frac{q}{2mc}mvr =$.

A Appendix

B Useful Links